

CO1 Assessment Tool 1

Analytical Problem Solving

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1. Water Jug Problem

Jugs: 4-gallon and 3-gallon

Initial state: (0,0)

Goal: 2 gallons in 4-gallon jug.

Let (x,y) = water in 4G and 3G jugs.

Step	Action	State
1	Fill 3G	(0,3)
2	3G → 4G	(3,0)
3	Fill 3G	(3,3)
4	3G → 4G	(4,2)
5	Empty 4G	(0,2)

6	3G → 4G	(2,0)
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Solution: $(0,0) \rightarrow (0,3) \rightarrow (3,0) \rightarrow (3,3) \rightarrow (4,2) \rightarrow (0,2) \rightarrow (2,0)$

2. Mars Rover as an Intelligent Agent

i. Percepts

- Camera/images
- Temperature and soil sensors
- Spectrometer
- Obstacle sensors
- Location and battery status

ii. Environment

- **Partially observable**
- **Stochastic**
- **Sequential**
- **Dynamic**
- **Continuous**
- **Single-agent**
- Partially unknown

iii. Actions

- Move and turn
- Avoid obstacles
- Capture images
- Collect and analyse samples

- Communicate with Earth
- Manage battery/power

iv. Performance

Measured by:

- Scientific discoveries
- Useful samples collected
- Area explored
- Analysis accuracy
- Energy efficiency
- Safety
- Successful communication

v. Suitable Agent

Model-based utility agent

It maintains an internal model because Mars is partially observable and chooses actions based on **scientific value, safety, energy and time**.

3. 8-Queens Problem

Problem

Place **8 queens on an 8×8 chessboard** so that no two queens attack each other.

Problem Formulation

Initial State: Empty board.

State: Arrangement of queens on the board.

Action: Place a queen in the next column if it does not conflict with existing queens.

Transition: Adding one queen creates a new state.

Goal Test: 8 queens placed with no common row, column or diagonal.

Path Cost: Number of queen placements.

State Representation

$[r_1, r_2, r_3, \dots, r_8]$

where r_i is the row of the queen in column i .

Search Strategy

DFS with Backtracking is suitable.

Place queen

↓

Valid?

/ \

Yes No

↓

↓

Next Backtrack

A heuristic can be:

$h(n)$ = number of attacking queen pairs

Goal: **$h(n) = 0$**

4. OLA Cab Booking Agent

Suitable Agent: Utility-Based Agent

The agent compares cab types using:

- Cost
- Comfort
- Travel time
- Availability

It selects the cab with the **highest utility** according to customer preference.

Pseudocode

BEGIN

Input source, destination, preference

Get available cabs

For each cab:

 Get cost, time, comfort, availability

 Calculate utility

Select cab with highest utility

Display cab, fare and time

If customer confirms:

 Book cab

Else:

 Cancel

END

Example: Cost-conscious $\rightarrow$ Shared; comfort-conscious $\rightarrow$ Prime.

5. Uniform Cost Search (UCS)

Given Graph

Edges:

- $S \rightarrow A = 1$
- $S \rightarrow G = 12$
- $A \rightarrow B = 3$
- $A \rightarrow C = 1$
- $B \rightarrow D = 3$
- $C \rightarrow D = 1$
- $C \rightarrow G = 2$
- $D \rightarrow G = 3$

UCS expands the node with the lowest cumulative cost.

Steps

1. Start S

A=1, G=12

Choose A(1).

2. Expand A

C=2, B=4, G=12

Choose C(2).

3. Expand C

D=3, G=4

Choose D(3).

4. Expand D

New path to G = $3+3=6$, but existing G = 4, so keep 4.

5. Choose G(4) → Goal reached.

Final Answer

Optimal Path:

S → A → C → G

Minimum Cost = $1 + 1 + 2 = 4$

UCS Expansion Order

S → A → C → D → G